

Predicting NFL Player Contract Values Using Position-Specific Regression Models

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Abstract — The National Football League (NFL) operates under a hard salary cap, making it essential for teams to accurately assess player value when negotiating contracts. While existing research highlights the relationship between payroll structure and team performance, there remains a gap in predictive modeling at the individual player level. This study introduces a position-specific regression framework to estimate future NFL contract values using performance metrics and demographic characteristics. Salary data will be collected from Spotrac, while on-field statistics will be sourced from Pro Football Reference. Separate models will be developed for quarterbacks, running backs, and wide receivers to ensure that predictor variables reflect the responsibilities and production metrics unique to each position. Model performance will be evaluated using R-squared and RMSE to compare predictive strength across groups. By identifying key performance indicators that drive financial outcomes, this project aims to support more informed contract negotiations for both teams and athletes while advancing the analytical understanding of market dynamics in professional sports. Ultimately, our findings will provide insight into how individual performance translates into compensation under the constraints of the league’s salary-cap environment.

Keywords—NFL, regression modeling, salary prediction, contract valuation, player performance

I. INTRODUCTION

Player contracts in the modern NFL are complex financial instruments that reflect not only on-field performance, but also market dynamics, salary-cap constraints, and long-term organizational strategy. Teams must optimize spending to remain competitive, while players and agents seek to maximize earnings. Predictive modeling of contract values therefore offers value to all stakeholders, improving negotiation leverage, salary transparency, and competitive balance.

Existing studies have examined aspects of salary determination but often focus on either team-level payroll effects or specific contract categories. For example, Susich (2024) [1] finds that wider salary distribution within a roster is associated with greater team success. Smith (2024) [2] predicts whether rookie quarterbacks will earn a second contract and estimates its fair value using GLMs and kernel regression. A

third study by Keefer and Kniesner (2024) [3] demonstrates peer-based inflation, showing that salaries rise when other players at the same position sign record-breaking deals. Together, these works establish that NFL compensation is shaped by performance, positional market trends, and comparison effects.

To integrate these dimensions, our study models contract valuation at the individual level using position-specific multiple linear regression. By tailoring predictors to each offensive role, we reduce statistical noise and more accurately quantify how performance contributes to contract outcomes. This approach aims to advance the analytics of player value while producing actionable salary projections for future negotiation cycles.

II. RELATED WORKS

Machine learning and statistical models have been applied across sports analytics, but relatively few studies target the prediction of NFL contract values at the individual level. Prior research focuses on:

- Salary distribution and team success: Wider salary dispersion can correlate with higher team win percentages. [1]
- Second contract modeling: Predictive models for quarterbacks’ second contracts incorporate performance and efficiency metrics. [2]
- Peer group effects: Compensation often rises when other comparable players reset the market at the same position. [3]

However, most previous work either examines team-level factors or analyzes narrow contract outcomes (e.g., rookies or elite-tier contracts). There is a lack of work modeling position-specific determinants of contract value, particularly using a consistent framework across all offensive and defensive groups.

This study addresses that gap by using standardized box-score statistics and demographic information to predict $\log(\text{contract value})$ within each role.

III. MODEL IMPLEMENTATION AND METHODOLOGY

A. Data Collection and Sources

The dataset used in this study will combine information from two primary sources: Spotrac and Pro Football Reference. Spotrac provides comprehensive player contract details, including annual salary, total contract value, signing bonuses, and guaranteed money. Pro Football Reference offers extensive performance statistics across positions and seasons. Data will cover players from the 2013 through 2024 NFL seasons to ensure recent relevance and consistent salary cap context.

Each player entry will include both quantitative and qualitative variables, encompassing demographic attributes (age, experience, draft round), contract attributes (total value, average annual value, guarantees), and performance statistics (yards, touchdowns, receptions, completion rate, and more). The dataset will be cleaned and standardized, with missing values addressed through imputation or removal depending on completeness. All monetary variables will be adjusted for inflation and salary cap growth to ensure comparability across years.

B. Preprocessing

Several preprocessing steps were applied to prepare the dataset for modeling. First, players were assigned to functional position groups, quarterbacks, running backs, wide receivers/tight ends, and defensive players, based on production patterns rather than roster labels to ensure that predictors accurately reflected each role's responsibilities. Continuous performance variables were standardized using z-scores to place them on a common scale, and the dependent variable, total contract value, was transformed using the natural logarithm to correct for right-skewed salary distributions. Observations containing incomplete or unusable contract or performance data were removed, while minor missingness was handled through median imputation within position groups. Finally, each dataset was partitioned into training (60%), testing (20%), and validation (20%) subsets using a reproducible random seed. This framework supports robust model evaluation and reduces the risk of overfitting.

C. Model Design

Separate regression models were developed for each offensive and defensive position group to account for the distinct statistical profiles and valuation patterns associated with each role. The dependent variable across all models was $\log(\text{total contract value})$, while independent variables included position-specific performance metrics and demographic factors such as age and experience.

For quarterbacks, the cleaned-up model has a total of 194 different players. The model incorporated passing yards, touchdowns, interceptions, completion percentage, and team win percentage. Running back models had a total of 346 different players. The model utilized rushing volume, rushing efficiency, and receiving involvement, whereas wide receiver/tight end models included receptions, receiving yards, yards after catch, catch percentage, and receiving touchdowns.

The wide receiver/tight end model had a total of Defensive models incorporated tackles, sacks, quarterback hits, and turnovers.

Although ridge regression and LASSO regression were tested to evaluate the stability of coefficient estimates, ordinary least squares ultimately provided sufficient interpretability and performance. Model accuracy was assessed using R-squared and root mean squared error (RMSE), enabling comparisons of predictive strength across position groups and identifying how strongly performance metrics explain variations in contract value.

IV. PERFORMANCE EVALUATION METRICS

Model performance will be evaluated using the following metrics: R-squared (R^2) to measure the proportion of variance in contract value explained by the model, Root Mean Squared Error (RMSE) to assess predictive error, and Adjusted R^2 for comparison between models with differing predictor counts. Comparative analysis across position-specific models will determine which roles exhibit the most predictable salary structures. A model with higher R^2 and lower RMSE will indicate that contract valuation in that position is more closely tied to measurable performance metrics.

V. EXPERIMENTAL SETUP AND IMPLEMENTATION

All modeling, preprocessing, and performance evaluation were conducted in R using the tidyverse ecosystem and a series of custom functions implemented within the project's RMarkdown workflow. Linear regression models were estimated using the `lm()` function, while user-defined helper functions were employed to generate reproducible training, testing, and validation splits for each position group. Additional utilities summarized model metrics, standardized predictors, and produced visualizations of coefficient estimates. Coefficient plots were created to display standardized effect sizes, enabling comparison of variable influence across offensive and defensive position groups.

The overall experimental procedure followed the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework. First, data understanding involved merging contract and performance data and identifying relevant predictors for each position. Second, data cleaning and preprocessing included handling missing values, standardizing variables, and transforming the contract value response. Third, a structured train/test/validation design was implemented to support robust model evaluation. Fourth, models were developed separately for quarterbacks, running backs, wide receivers/tight ends, and defensive players. Fifth, model performance was assessed using R-squared and root mean squared error (RMSE) on both test and validation sets. Finally, results were interpreted through coefficient analysis and comparative evaluation to determine which performance metrics most strongly influence contract valuation within and across positions.

VI. RESULTS AND DISCUSSION

A. Model Performance Across Groups



Across all offensive and defensive models, quarterback salaries demonstrated the strongest relationship with performance metrics. QB models achieved the highest R^2 values and the lowest RMSE on test and validation data. Passing yardage, touchdowns, and completion percentage were among the strongest positive predictors of salary, while interceptions contributed negatively.

The RB model performed moderately well but exhibited more variability. Rushing yardage, rushing attempts, and receiving usage all demonstrated positive associations with contract value, indicating that versatile multi-phase backs are more valuable in the current NFL market.

The WR/TE model consistently showed the weakest performance, with low R^2 and higher RMSE. This suggests that box-score receiving statistics alone do not fully explain pass-catcher salaries, which may depend on film-based metrics (route separation, contested-catch ability), athletic traits, and market timing.

The defensive model performed in a mid-tier range. Sacks and quarterback hits were meaningful predictors, reflecting that pass-rushing skill drives defensive salaries. Tackles and interceptions showed weaker effects, possibly due to position-dependent variability.

B. Coefficient Interpretation

Offense

Quarterbacks showed strong positive coefficients for passing yards, touchdowns, and completion percentage, indicating that consistent production and efficiency drive higher salaries. Interceptions had a negative coefficient, aligning with economic expectations that turnover-prone QBs are less valued.

For running backs, rushing yards and attempts were significant predictors, but receiving yards and receptions were also important, reinforcing the modern preference for dual-threat backs.

Pass-catchers displayed weaker and more inconsistent coefficient patterns, reflecting a mismatch between box-score statistics and contract valuation. This aligns with league trends in which WR/TE contracts increasingly depend on athleticism, route-versatility, and scheme-specific traits.

Defense

Defensive coefficients showed that sacks and quarterback hits were among the strongest predictors of salary across defenders. Tackles had a smaller positive effect, while interceptions were variable. This supports the notion that pass-rushing, one of the most scarce and valuable defensive skills, drives the market more than coverage or tackling volume.

VII. CONCLUSION AND FUTURE WORK

A. Conclusion

The results of this study demonstrate clear differences in how effectively performance metrics explain contract valuation across NFL position groups. The quarterback model displayed the strongest predictive performance, achieving an R^2 of 0.688 on the test set and 0.577 on the validation set, with RMSE values of 0.893 and 0.944, respectively. These results confirm that quarterback salaries are the most directly linked to measurable production, particularly passing yardage, touchdown efficiency, and completion percentage. The consistency across test and validation sets indicates a relatively stable relationship between quarterback performance and salary outcomes in the modern NFL.

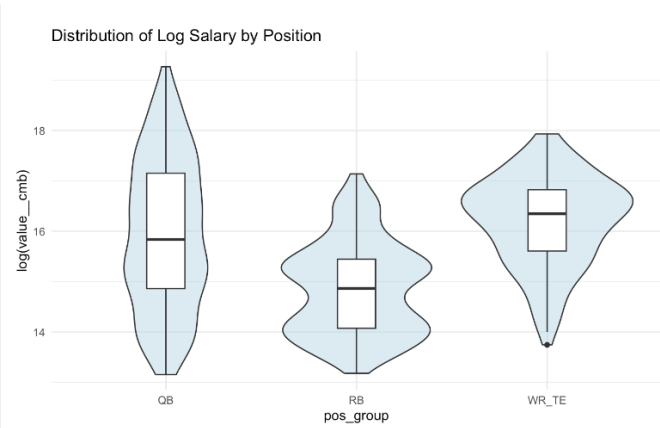
Running backs demonstrated moderate predictive capability, with test-set $R^2 = 0.268$ and validation $R^2 = 0.381$, accompanied by RMSE values of 0.790 and 0.740. These metrics suggest that core rushing statistics, combined with receiving involvement, explain a meaningful portion of running back contract variability, although the relationship is weaker and less consistent than for quarterbacks. This aligns with league-wide uncertainty in the running back market and the reduced financial emphasis placed on rushing volume alone.

Wide receivers and tight ends showed the weakest model performance, with negative R^2 values in both the test (-0.186) and validation (-0.178) sets. Despite RMSE values near 0.997 and 0.857, the negative R^2 indicates that the WR/TE model performs worse than a simple baseline prediction. This suggests that box-score receiving metrics alone, such as receptions, yards, and touchdowns, largely fail to capture how

pass-catchers are valued in contract negotiations. This is consistent with the reality that WR and TE salaries are heavily influenced by non-quantified traits such as route-running ability, separation, contested-catch ability, athletic profile, team scheme, and market timing effects.

The defensive model demonstrated moderate predictive strength, with test and validation R^2 values of 0.430 and 0.496, and RMSE values of 0.860 and 0.821. These results show that defensive salary outcomes are partially explained by measurable performance, particularly pass-rush statistics such as sacks and quarterback hits. Still, substantial unexplained variance remains consistent with the complexity of evaluating defensive performance using only box-score metrics.

Overall, the findings validate the importance of position-specific modeling, reveal strong predictive relationships for quarterbacks, moderate relationships for running backs and defensive players, and show that receiving roles require additional contextual information beyond standard statistics. These results highlight the limitations of traditional performance measures in fully capturing market value for certain positions.



B. Future Work

Future work should extend this modeling framework in several meaningful directions. First, while offensive players

were successfully separated into QBs, RBs, and WR/TE categories, the defensive model currently aggregates all defenders into a single group. A promising improvement would involve splitting defensive players into functional subgroups such as edge rushers, interior linemen, linebackers, and defensive backs to better capture the unique statistical profiles and market dynamics within each defensive role. Such granularity would likely improve explanatory power and reveal role-specific salary drivers.

Second, the present study relies primarily on box-score performance data. Significant gains in predictive accuracy may be achievable by incorporating external and contextual factors that strongly influence contract valuation. Examples include market value trends at each position, timing of major extensions, player injury history, team cap space at the time of signing, and advanced analytics such as EPA/play, PFF grades, route separation metrics, or tracking data. These variables likely account for a substantial portion of the unexplained variance observed in WR/TE and some defensive positions.

Finally, future research could explore more sophisticated machine-learning approaches such as gradient boosting, random forests, elastic net modeling, or neural network architectures to capture nonlinear relationships and interactions that linear regression may not detect. Incorporating these enhancements would further advance the predictive modeling of NFL contract values and offer greater insight into the complex dynamics governing salary determination in professional football.

REFERENCES

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